

SIMON OKOSODO

Data Scientist | Machine Learning Engineer — Judgment-Led Validation, 2x Precision@50 Lift (0.400→0.820) on 9.8M Rows, Deployed Live

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PROFESSIONAL SUMMARY

Data Scientist / Machine Learning Engineer whose value is judgment, not just execution: I decide which calls stay permanently human, which claims the evidence can and cannot carry, and which number is honest versus merely flattering — before a model ever ships. Designed, trained, validated, and deployed a client-grouped Random Forest ranking model lifting content-review Precision@50 from a 0.400 rule-based baseline to 0.820 on 9.8 million rows of real production search data, after first catching that a naive random split scored a suspicious 1.000 and would have quietly inflated the result. Freed roughly 21 editor-hours weekly at the same headcount, shipped as a live, key-authenticated REST API with an MCP-compatible agent tool. Background in process-driven, zero-non-compliance quality control forged the same evidence-first discipline now applied to every model I validate: AI can generate code and candidate results faster than a team can verify them — I close that gap by making certain decisions non-discretionary and testing every result locally before it ships, no exceptions.

SKILLS

Judgment & Decision Discipline: Framing decisions before modeling, cost-of-error analysis, leakage-hunting (deliberate leak-plant-and-remove), honest claim capping (observed vs. proven), human-only no-go lists

Languages & Libraries: Python, pandas, scikit-learn, NumPy, joblib, SQL

Data & Querying: SQL, DuckDB (remote Parquet at 79M-row scale), Hugging Face Datasets

Modeling & Validation: Logistic Regression, Random Forest, Gradient Boosting, Precision@K, client-grouped & time-aware validation, data contracts

Deployment & Tooling: FastAPI, REST API design, key-based auth, Render, Docker, model versioning, Git/GitHub, GitHub Pages, Google Colab, Claude (Projects, API, Agent Skills, MCP), ChatGPT

Communication: Technical/research writing, stakeholder documentation, prompt engineering

EXPERIENCE

Data Science Intern — FlyRank (Remote)

Jul 2026 – Sep 2026

- Framed the decision before touching data: named who acts on the model's output and the cost of a wrong call, then wrote and verified a data contract against the live 79M-row production warehouse (DuckDB over Hugging Face Parquet), confirming grain with zero violations rather than assuming it.
- Chose to prove a simple rule wasn't enough before reaching for a model: engineered a 5-feature, leakage-safe, forward-looking (Feb→Mar) training set from 130,969 eligible rows across 55 clients and 331,437 content items; deliberately planted a known leak, watched accuracy spike from 0.983 to 1.000, then removed it — proving the leak-detection discipline before trusting it on the real model.
- Made the harder validation call over the easier one: trained a Random Forest under a client-grouped split (no client's pages in both train/test) specifically to rule out memorization, lifting Precision@50 from 0.400 to 0.820 (2.05x) — and reported a naive random-split score of 1.000 alongside it as a memorization warning, not a result, because the honest number mattered more than the impressive one.
- Caught and corrected my own errors before they reached a final claim: found a run-to-run reproducibility bug and a hardcoded baseline mismatch, recomputing the editor-hours-freed figure from an inflated ~29 to the verified ~21/week.
- Built a 4-category diagnosis layer (genuine decline / SERP-answered / CTR-fixable / stable) mapped to specific editor actions, plus an explicit no-go list of calls that stay human-only; found 10.5% of pages showed a SERP-answered pattern invisible to the old rule.
- Verified production matched local inference exactly before calling it done: deployed the model as a live, key-authenticated REST API (FastAPI, Docker, Render) with an MCP-compatible agent tool, and published a reproducible capstone research paper, capping every claim to what the evidence actually supports.

ADDITIONAL PROFESSIONAL EXPERIENCE

Chief Chemist, QC Laboratory — Oando Plc

Oct 2018 – Mar 2019 · Lagos, Nigeria

- Applied a B.Eng. in Chemical Engineering to lead quality control testing (density, flash point, distillation, water content) for fuel shipments against ASTM and JIG international standards, holding a zero non-compliance record across every shipment tested — the same evidence-first, judgment-under-consequence discipline now shapes how I validate a model before I stand behind its number.

Sales & Service Operations — Nigerian Breweries Plc

Nov 2020 – May 2022 · Lagos, Nigeria

- Tracked and improved KPI performance across 5 distributors and ~220 retail outlets using daily call-fulfillment, strike-rate, and volume data; raised Net Numeric Distribution from ~60% to ~78% and outlet coverage from 150 to 200 of 220 through data-driven route and priority decisions.

Fuel Depot Operative — TotalEnergies

Dec 2011 – Dec 2014 · Lagos, Nigeria

- Coordinated high-throughput loading operations (20–50 trucks/shift, up to 1.65M litres/shift) and maintained calibration integrity of metering and storage systems, sustaining a zero safety-incident record throughout tenure.

PROJECTS

Diagnosis-First Content Triage — FlyRank ML Capstone

Owned every decision end to end: research framing, data contract, leakage audit, transparent baseline, feature engineering, client-grouped validation, action playbook, public paper, and a live production API. Result: Precision@50 0.400→0.820 (2.05x), ~21 editor-hours/week freed.

[GitHub Repo](#) · [Live API](#) · [Research Paper](#)

PSO-ML20 — 20-Phase Machine Learning Framework (Independent)

Built a reusable ML lifecycle framework (data integrity, Deep Feature Synthesis, model tournament, leakage stress-testing) so every future project inherits the same judgment discipline by default; applied to a real-estate valuation model (King County dataset, $R^2 \sim 0.88$, MAPE ~10.7%) deployed as a live Streamlit app, and to a self-serve churn-prediction interface.

[GitHub Repo](#) · [Live App](#)

FlyRank Priority Digest — Personal AI Agent (Claude Projects)

Built an agent grounded in a real scored-queue CSV, explicitly constrained to refuse inventing an answer it cannot support; passed 3 documented eval cases, including a guardrail test confirming it declines rather than fabricates data for an unknown page ID.

EDUCATION

B.Eng., Chemical Engineering — University of Benin (UNIBEN), Nigeria. Coursework: process optimization, numerical methods, systems engineering.

Data Science and Machine Learning Certificate Program — May 2026, Tech-Agon Machine Learning Institute, Ghana.

CERTIFICATIONS

Anthropic (15): AI Fluency (Students, Educators, Small Business, Nonprofits, Builders) · Teaching the AI Fluency Framework · AI Capabilities & Limitations · Claude 101 · Claude Platform 101 · Claude Code 101 & In Action · Intro to Subagents · Intro to Agent Skills · Intro to MCP · MCP: Advanced Topics · Claude with the Anthropic API · Claude with Amazon Bedrock

IBM SkillsBuild (3): Foundations in Generative AI · Introduction to Retrieval-Augmented Generation · Prompt Engineering: Shaping Better AI Responses

Google x Kaggle: 5-Day AI Agents: Intensive Vibe Coding Course (Jul 2026)